

Page 5: Project Development Phase

1. Introduction

The **Project Development Phase** details the technical execution, coding standards, component implementation, and integration strategy for **Smart Budget AI**. This phase translates design wireframes and architectural blueprints into a fully functional software product.

2. Development Methodology & Setup

- **Agile Scrum Methodology:** Development is structured into 2-week sprint cycles with daily standups, code reviews, and sprint retrospectives to ensure continuous integration and iterative progress.
- **Version Control System:** Managed via Git with feature-branch workflows (GitFlow) hosted on GitHub/GitLab.
- **CI/CD Pipeline:** GitHub Actions / GitLab CI for automated linting, unit testing, and continuous deployment to staging servers.

3. Module-Wise Implementation Strategy

3.1 Frontend Development

- **Cross-Platform Interface (React Native / Flutter):** Developed reusable components for real-time dashboards, transaction lists, interactive charts, and user profiles.
- **State Management:** Utilized Redux Toolkit / Provider for predictable state handling across screens.
- **API Integration:** Implemented Axios / Fetch clients with interceptors to handle OAuth 2.0 access tokens, request timeouts, and error handling.

3.2 Backend & API Infrastructure

- **RESTful & WebSocket APIs (Python FastAPI / Node.js):** Built secure REST endpoints for CRUD operations on transactions, accounts, and user budgets. WebSocket connections enable real-time notifications for budget alerts.
- **Database & Caching Layer:** Provisioned PostgreSQL with indexed queries for fast transaction retrieval, alongside Redis caching for active user sessions and frequent analytic queries.
- **Authentication & Security:** Implemented JWT (JSON Web Tokens) with short expiry times, password hashing using bcrypt, and request validation to prevent SQL injection and XSS attacks.

3.3 AI & Machine Learning Core Pipeline

- **OCR Receipt Processing Module:** Built a image preprocessing pipeline (grayscale,

contrast adjustment, deskewing) integrated with Tesseract OCR / EasyOCR to parse structured merchant, date, and total amount data from receipt images.

- **NLP Expense Categorization Model:** Trained a text classification model (using Scikit-learn / Hugging Face Transformers) to automatically map transaction descriptions to categories (e.g., "Starbucks" → *Food & Dining*).
- **Predictive Forecasting Engine:** Implemented time-series forecasting models (ARIMA / Prophet) to analyze historical spending patterns and project end-of-month expenditure.
- **Conversational Financial Assistant:** Integrated an LLM-driven module with custom prompt engineering to process natural language user queries safely.

4. Code Structure & Repository Organization

```
smart-budget-ai/
├── frontend/                # React Native / Flutter Application
│   ├── src/assets/         # Fonts, Images, Icons
│   ├── src/components/     # Reusable UI Widgets
│   ├── src/screens/        # Dashboard, Analytics, Chat UI
│   └── src/services/       # API Services & State Management
├── backend/                # Server & API Engine
│   ├── app/api/            # REST API Endpoints & Routes
│   ├── app/core/           # Auth, Security & Middleware
│   ├── app/models/         # Database Models & Schemas
│   └── app/services/       # Business Logic & Database Queries
├── ai_engine/              # Machine Learning Pipeline
│   ├── models/             # Saved ML / NLP Classifier Weights
│   ├── ocr/                # Receipt Processing Scripts
│   └── forecaster/         # Time-Series Forecasting Scripts
└── docs/                  # API Specifications & Setup Guides
```

5. Key Milestones Completed in Development

1. **Database & Auth Milestone:** Secure user registration, login, and database schema deployment.
2. **Core Ledger Milestone:** Full CRUD implementation for manual and automated transaction tracking.
3. **AI Pipeline Milestone:** Operational receipt scanner and auto-categorization model integration.
4. **Interactive UI Milestone:** Real-time dashboard with dynamic spending charts and conversational chatbot interface.